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Project Description

(Class Diagram at the end of file)

Abstract Classes

EventAbstract Class

This abstract base class provides common functionality for event-related classes.

Fields:

- `is_active`: Boolean field indicating if the event is currently active
- `is_deleted`: Boolean field indicating if the event has been marked for deletion
- `created_at`: DateTime field recording when the event was created
- `updated_at`: DateTime field recording when the event was last modified

TaskAbstract Class

This abstract base class provides common functionality for task-related classes.

Fields:

- `is_active`: Boolean field indicating if the task is currently active
- `is_deleted`: Boolean field indicating if the task has been marked for deletion
- `created_at`: DateTime field recording when the task was created
- `updated_at`: DateTime field recording when the task was last modified

User Management

User Class

Represents a user in the system, extending Django's `AbstractBaseUser`.

Functions:

1. `set_password(password)`
 - Description: Sets and encrypts the user's password
 - Parameters: `password` (str) - The plaintext password to encrypt
 - Returns: None
2. `str()`
 - Description: Returns string representation of the user

- Parameters: None
- Returns: User's email address

Fields:

- email: String field storing user's email address (used as username)
- is_active: Boolean field indicating if the account is active
- date_joined: DateTime field recording when the user joined
- last_updated: DateTime field recording last profile update

UserManager Class

Manages user creation and authentication operations.

Functions:

1. `_create_user(email, password)`
 - Description: Creates a new user with the given email and password
 - Parameters:
 - email (str) - User's email address
 - password (str) - User's password
 - Returns: New User instance

Event Management

Event Class

Represents calendar events in the system.

Functions:

1. `get_absolute_url()`
 - Description: Generates URL for viewing event details
 - Parameters: None
 - Returns: URL string for event detail page
2. `get_html_url()`
 - Description: Creates HTML link for event in calendar view
 - Parameters: None
 - Returns: HTML string containing event link
3. `str()`
 - Description: Returns string representation of the event
 - Parameters: None
 - Returns: Event title

Fields:

- title: String field storing event title
- description: Text field storing event description
- start_time: DateTime field for event start
- end_time: DateTime field for event end

- user: Foreign key to User model

EventManager Class

Manages event retrieval and filtering operations.

Functions:

1. `get_all_events(user)`
 - Description: Retrieves all active events for a user
 - Parameters: user (User) - User whose events to retrieve
 - Returns: QuerySet of Event objects
2. `get_today_events(user)`
 - Description: Retrieves events scheduled for today
 - Parameters: user (User) - User whose events to retrieve
 - Returns: QuerySet of Event objects
3. `get_completed_events(user)`
 - Description: Retrieves past events
 - Parameters: user (User) - User whose events to retrieve
 - Returns: QuerySet of Event objects
4. `get_upcoming_events(user)`
 - Description: Retrieves future events
 - Parameters: user (User) - User whose events to retrieve
 - Returns: QuerySet of Event objects

Task Management

Task Class

Represents tasks and assignments in the system.

Functions:

1. `get_absolute_url()`
 - Description: Generates URL for viewing task details
 - Parameters: None
 - Returns: URL string for task detail page
2. `to_dict()`
 - Description: Converts task instance to dictionary format
 - Parameters: None
 - Returns: Dictionary containing task data
3. `str()`
 - Description: Returns string representation of the task
 - Parameters: None
 - Returns: Task title

Fields:

- `is_completed`: Boolean field indicating task completion status

- title: String field storing task title
- description: Text field storing task description
- deadline: DateTime field for task due date
- user: Foreign key to User model

TaskManager Class

Manages task retrieval and filtering operations.

Functions:

1. `get_all_tasks(user)`
 - Description: Retrieves all active tasks for a user
 - Parameters: user (User) - User whose tasks to retrieve
 - Returns: QuerySet of Task objects
2. `get_today_tasks(user)`
 - Description: Retrieves tasks due today
 - Parameters: user (User) - User whose tasks to retrieve
 - Returns: QuerySet of Task objects
3. `get_upcoming_tasks(user)`
 - Description: Retrieves future tasks
 - Parameters: user (User) - User whose tasks to retrieve
 - Returns: QuerySet of Task objects

External Service Integration

FileProcessor Class

Handles processing of external data files and synchronization.

Functions:

1. `process_blackboard(username, password)`
 - Description: Processes Blackboard assignments into tasks
 - Parameters:
 - username (str) - Blackboard username
 - password (str) - Blackboard password
 - Returns: Dictionary containing processing results
2. `process_edusoft_timetable(username)`
 - Description: Processes Edusoft schedule into events
 - Parameters:
 - username (str) - Edusoft username
 - Returns: Dictionary containing processing results
3. `clear_event_data()`
 - Description: Removes existing events before sync
 - Parameters: None
 - Returns: Dictionary containing deletion results
4. `clear_task_data()`

- Description: Removes existing tasks before sync
 - Parameters: None
 - Returns: Dictionary containing deletion results
5. `__process_timetable(df)`
 - Description: Processes timetable data into standardized format
 - Parameters: `df` (DataFrame) - Raw timetable data
 - Returns: List of processed course data
 6. `__create_recurring_events(course_data)`
 - Description: Creates recurring events for courses
 - Parameters: `course_data` (dict) - Course schedule information
 - Returns: Number of events created

SyncService Class

Orchestrates synchronization between external services and the application.

Functions:

1. `sync_edusoft_schedule(user, username, password)`
 - Description: Synchronizes Edusoft class schedule
 - Parameters:
 - `user` (User) - User instance
 - `username` (str) - Edusoft username
 - `password` (str) - Edusoft password
 - Returns: Dictionary containing sync results
2. `sync_blackboard_tasks(user, username, password)`
 - Description: Synchronizes Blackboard assignments
 - Parameters:
 - `user` (User) - User instance
 - `username` (str) - Blackboard username
 - `password` (str) - Blackboard password
 - Returns: Dictionary containing sync results

